

The Idiosyncratic Channel: Systematic Cancellation in Conditional BAB and a Necessary-and-Sufficient Characterization

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ABSTRACT

No standard asset pricing model generates a negative relationship between aggregate volatility and the betting-against-beta (BAB) return. The BAB formula divides each asset's expected return by its beta, annihilating every component of expected returns proportional to beta, including the entire systematic risk premium. The key structural insight (Theorem 1) is that the only term surviving beta-division is the idiosyncratic variance bracket. Among standard portfolio constraints, only leverage constraints actively push BAB in the wrong direction: as they tighten with volatility, they raise BAB. Variance-budget, tracking-error, beta-budget, and VaR constraints are direction-neutral.

The resolution requires a channel that survives beta-division. The only such channel is idiosyncratic volatility. The paper derives an exact necessary-and-sufficient threshold condition for $d\text{BAB}/d\sigma_m < 0$ in terms of the relative idiosyncratic-volatility sensitivities of high- versus low-beta stocks. Under the

OLS residual definition of idiosyncratic risk, equity-as-call-option curvature is misattributed as idiosyncratic variance, and the misattribution is larger for more levered (high-beta) firms — so the threshold is satisfied as a structural equilibrium outcome (Proposition OLS-MF).

Using 3.37 million stock-months from CRSP (July 1963 to December 2024, 26,122 unique stocks), estimates of the idiosyncratic-volatility sensitivity from rolling regressions confirm the characterization condition across all five beta quintiles. The estimated sensitivity rises monotonically from 0.24 (Q1) to 1.33 (Q5); the empirical ratio of 5.48 exceeds the theoretical threshold of 2.97 by a factor of 1.85. The BAB- $\Delta\sigma$ regression yields slope -0.87 ($t = -2.11$). At $\Theta_0 = 3$, the model accounts for 55% of the empirical slope; a complete quantitative account requires additional mechanisms. When $\sigma_m(t)$ follows a mean-reverting square-root process, Itô's lemma applied to $BAB(\sigma_m(t))$ delivers a timing proposition: BAB is expected to recover following a volatility spike, providing structural foundations for volatility-managed BAB strategies.

PROVENANCE

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